Learning Objective 4.1: I can describe FSK modulation and implement an FSK transmitter and receiver in GNU Radio.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Transmitter settings.** A flowgraph samples at $f_s = 48$ kHz and sends FSK at 1200 baud with a deviation of $\Delta f = 600$ Hz, one bit per symbol.

   (a) Find the samples per symbol to enter in the Repeat block.

   (b) Find the sensitivity to enter in the Frequency Mod block.

   (c) Find the modulation index and estimate the bandwidth with Carson's rule.

   h =

   B =

   (d) A classmate builds the same transmitter from two Signal Sources at $+600$ and -600 Hz and a selector that switches between them at each bit. Explain why Frequency Mod gives a more compact spectrum.

2. **Receiver settings.** The receiver for the link of Question 1 follows the L8 flowgraph: Low Pass Filter, Quadrature Demod, Moving Average, Symbol Sync, and Binary Slicer.

   (a) Find the Quadrature Demod gain that puts the two tones at $+1$ and -1.

   (b) Find the low-pass filter cutoff that passes the Carson bandwidth.

   (c) A classmate leaves the Quadrature Demod gain at 1. Find the levels the two tones produce, and explain whether the slicer still decides correctly.

   (d) The classmate also sets the filter cutoff to 8 kHz. Explain what changes on the eye diagram, and what happens to the noise voltage at which the text first breaks.

3. **Noise.** The L8 link samples at 32 kHz, sends 1000 baud with $\Delta f = 1$ kHz, and filters to 4 kHz. Its Channel Model has a noise voltage slider. At 0.5 the two tones stand about 25 dB above the noise floor on the spectrum, the eye is wide open, and every line arrives.

   (a) The noise slider moves from 0.5 to 1.3. Describe what you expect to see on the spectrum, on the eye diagram, and in the received text.

   (b) A spike crosses zero on the eye diagram, but away from the sampling instant. Explain whether it causes a bit error.

   (c) The link doubles its symbol rate to 2000 baud and keeps $\Delta f = 1$ kHz. Find the new bandwidth and filter cutoff, and explain whether the text breaks at a higher or a lower noise setting than before.

   (d) On the L8 link, the text goes from clean to almost no packets between noise voltages of about 1.0 and 2.0. Lesson 2 designs a link with margin. Explain, from that observation, why margin matters for FSK.

4. **An FSK remote on a spectrogram.** An RTL-SDR capture, viewed as a spectrogram in URH, shows a burst with two tracks, at 433.900 and 433.940 MHz. In the alternating preamble, the signal stays on each track for 0.417 ms.

   (a) Find the center frequency, the deviation, and the symbol rate.

   f_c =

   Δf =

   R_s =

   (b) The remote's crystal is accurate to 10 ppm. Find the largest frequency offset at 433.92 MHz.

   (c) An FSK receiver tolerates an offset of about half its deviation. Explain whether a 4.3 kHz offset breaks the remote's link, and whether it would break the L8 link if that link ran at 433.92 MHz.

   (d) With a 600 Hz offset the L8 demodulator outputs levels near $+1.6$ and -0.4. Name one change to the receiver that would put them back at ± 1.

Documentation: _____